特殊函数系列课程
Lambert W function

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Lambert W function

- **Lambert 生平简介:**
  Johann Heinrich Lambert (1728–1777)
  Swiss polymath
  Made important contributions to the subjects of mathematics, physics (particularly optics), philosophy, astronomy and map projections.

目录 I

1 一、历史来源

2 二、定义

3 三、性质
- 1. 分类
- 2. 图像
- 3. 导数和积分
- 4. 级数和恒等式

4 四、应用

Lambert W function 历史来源

在1758年，Lambert为了求解三项式方程 (trinomial equation):

$$x = q + x^m, \tag{1}$$

利用了级数展开的方法。后来 Euler 作了进一步发展，引入代换:

$$x \to x^{-\beta}, \quad m \to \alpha/\beta, \quad q \to (\alpha - \beta)v,$$

将原方程 (1)改写为一个更加对称的表达式

$$x^\alpha - x^\beta = (\alpha - \beta)vx^{\alpha+\beta}. \tag{2}$$

Lambert W function 历史来源

于是原来Lambert版本的级数展开表达式，Euler改写成下面的形式，其中 n 是整数

$$\begin{aligned} x^n &= 1+nv+\frac{1}{2}n(n+\alpha+\beta)v^2 \\ &+\frac{1}{6}n(n+\alpha+2\beta)(n+2\alpha+\beta)v^3 \\ &+\frac{1}{24}n(n+\alpha+3\beta)(n+2\alpha+2\beta)(n+3\alpha+\beta)v^4 \\ &+etc. \end{aligned} \tag{3}$$

R.M. Corless, et al.,*On the Lambert W function*, Adv. Comput. Math. **5** (1996) 329.

Lambert W function 历史来源

现在考虑一个特殊情形：把对称的三项式方程(2)式写为 $(x^\alpha - x^\beta)/(\alpha - \beta) = vx^{\alpha+\beta}$，再令$\alpha = \beta$，则有[1]

$$\ln x = vx^\alpha.$$

Euler 注意到，只要能够得到 $\alpha = 1$ 的解，那么相应地可以得到任意 $\alpha \neq 0$ 的解，简要证明如下：

[1]这里用到了极限表达式：$\lim_{n\to 0}(a^n - 1)/n = \ln a$.

Lambert W function 历史来源

证明：
在方程 $\ln x = vx^{\alpha}$ 两边同乘以 α，可以得到 $\ln x^{\alpha} = ux^{\alpha}$，此处 $u = \alpha v$，进而做代换 $y = x^{\alpha}$，则有 $\ln y = uy$，即对应于原来方程中 $\alpha = 1$ 的情形。■

Lambert W function 历史来源

为了进一步求解所化简得到的方程，利用 Euler版本的级数表达式 (3)，将其改写为

$$\frac{x^n-1}{n}=v+\frac{1}{2}(n+\alpha+\beta)v^2+\cdots,$$

让 $\alpha=\beta=1$，然后令 $n=0$，于是得到

$$\ln x=v+\frac{2}{2!}v^2+\frac{3^2}{3!}v^3+\frac{4^3}{4!}v^4+\frac{5^4}{5!}v^5+etc.$$

因此，这个关于 v 的级数表达式记作 $-W(-v)$，也即引出了 Lambert W function $W(x)$.

Lambert W function 定义

Lambert W function, 也称为 Omega function 或者 product logarithm, 它来源于函数 $f(z) = ze^z$ 的反函数

- $z = f^{-1}(ze^z) = W(ze^z)$，该式两边同乘以 e^z，做代换 $z' = ze^z$，即有 $z' = W(z')e^{W(z')}$.

定义表达式：

$$z = W(z)e^{W(z)}, \quad \text{对于任意复数} z.$$

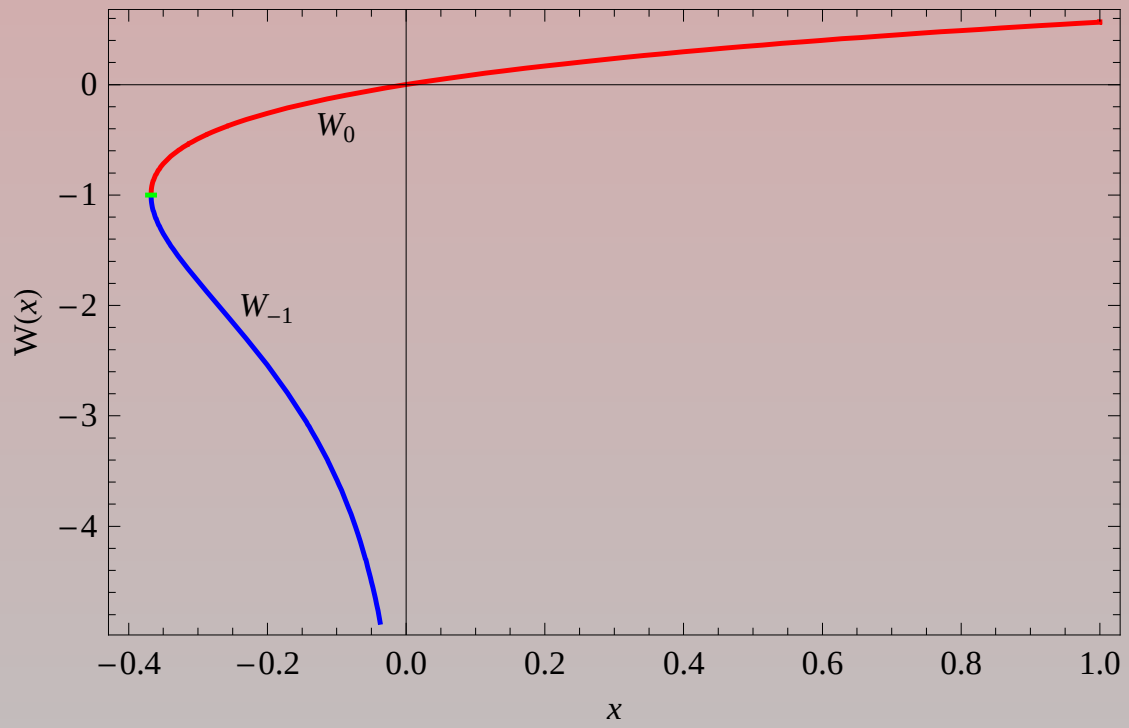
Lambert W function 分类

分类

- $W(z)$ 是一个多值函数，有无穷多个分支，记为 $W_k(z)$, $(k = 0, \pm1, \pm2, \pm3, \pm4, \cdots)$，而且每个分支都是一个单值函数
- 在实数范围内，即 $z = x$，有 $W_{-1}(x)$ 和 $W_0(x)$ 两支，其中 $W_0(x)$ 称为其主支：
  - 定义域：$x \geq -\frac{1}{e}$
  - 当 $W(x) \geq -1$，记为 $W_0(x)$ 或 $W(x)$，是单值函数
  - 当 $W(x) \leq -1$，记为 $W_{-1}(x)$，是单值函数
  - 在区间 $\left(-\frac{1}{e}, 0\right)$ 之间，是双值函数

Lambert W function 图像

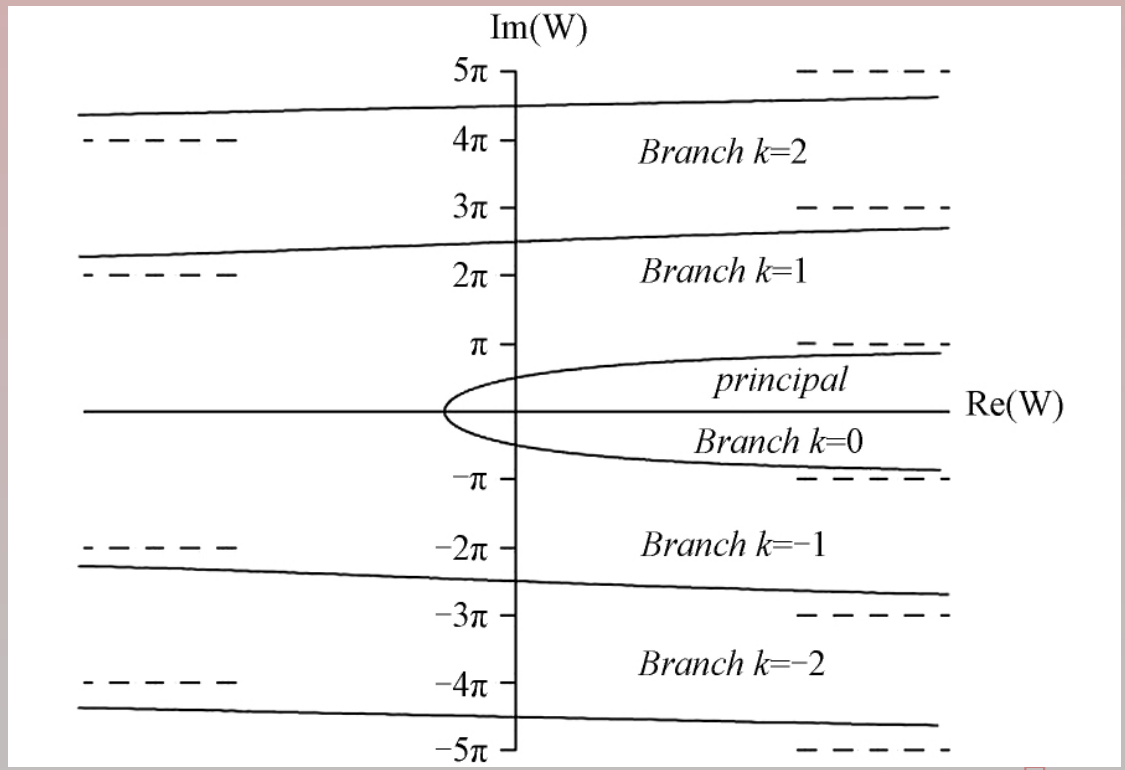
1. 关于 $W_{-1}(x)$ 和 $W_0(x)$ 的图像



Lambert W function 图像

2. 关于 $W(z)$ 各个分支的图像

R.M. Corless, et al.,*On the Lambert W function*, Adv. Comput. Math. **5** (1996) 329.



Lambert W function 导数

将方程 $x = W(x)e^{W(x)}$ 对 x 求导数，得到：

$$\frac{dW(x)}{dx} = \frac{1}{e^{W(x)}(1+W(x))} = \frac{W(x)}{x(1+W(x))} \quad (x \neq 0)$$

Lambert W function 导数

- n 阶导数：

$$\frac{d^n W(x)}{dx^n} = \frac{e^{-nW(x)} p_n(W(x))}{(1+W(x))^{2n-1}}$$

其中多项式 $p_n(W)$ 满足递推关系式 $p_{n+1}(W) = -(nW+3n-1)p_n(W) + (1+W)p_n'(W)$。当 $n \geq 1$ 时，多项式的初值 $p_1(W) = 1$，且有 $p_n(0) = (-n)^{n-1}$.

Lambert W function 积分

积分 I

$$\int W(x)dx = xW(x) - x + e^{W(x)} + C$$

推论

$$\int_0^e W(x)dx = e - 1$$

Lambert W function 积分

证明：令 $u = W(x)$，相应的根据 Lambert W function 的定义，可以得到 $x = ue^u$，因此

$$\begin{aligned}
\int W(x)dx &= xW(x) - \int xdW(x) = xW(x) - \int ue^u du \\
&= xW(x) - \int ude^u \\
&= xW(x) - \left(ue^u - \int e^u du\right) \\
&= xW(x) - ue^u + e^u + C \\
&= xW(x) - x + e^{W(x)} + C.
\end{aligned}$$

■

Lambert W function 积分

积分 II

$$\int xW(x)dx = \frac{1}{8}(2W(x)-1)(2W^2(x)+1)e^{2W(x)} + C$$

证明：令 $u = W(x)$，于是 $x = ue^u$，因此

$$\begin{aligned} \text{原式} &= \int ue^u \cdot u \cdot (u+1)e^u du = (3\text{次分部积分}) \\ &= \frac{1}{8}(2u-1)(2u^2+1)e^{2u} + C \end{aligned}$$ ■

Lambert W function 级数

级数[a]

[a] 见 Lambert W function 历史来源的内容

$$W(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n$$

收敛半径：$r = \lim_{k\to\infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k\to\infty} \frac{1+1/k}{(1+1/k)^k} = \frac{1}{e}$

Lambert W function 恒等式

特殊值

- $W(0) = 0 \qquad W(e) = 1 \qquad W(-1/e) = -1$
- $W'(0) = 1$
- $W\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}i$
- $W\left(-\frac{\ln a}{a}\right) = -\ln a, \qquad \frac{1}{e} \le a \le e$
- $W(1) = \Omega \approx 0.56714329$ (Omega constant)

Lambert W function 恒等式

恒等式

$$\ln W(x) = \ln x - W(x),\ (x > 0)$$

$$e^{nW(x)} = \left(\frac{x}{W(x)}\right)^n$$

Lambert W function 恒等式

递推关系 I： $x > 0, \quad W_0(x) > 0$

$$\begin{aligned} W_0(x) &= \ln x - \ln W_0(x) \\ &= \ln x - \ln(\ln x - \ln(\ln x - \cdots)) \\ &= \ln \frac{x}{\ln \frac{x}{\ln \frac{x}{\cdots}}} \end{aligned}$$

Lambert W function 恒等式

递推关系 II：$x < 0, \quad W_{-1}(x) < 0$

$$
\begin{aligned}
W_{-1}(x) &= \ln(-x) - \ln(-W_{-1}(x)) \\
&= \ln(-x) - \ln(-(\ln(-x) - \ln(-(\ln(-x) - \cdots)))) \\
&= \ln \frac{-x}{-\ln \frac{-x}{-\ln \frac{-x}{\cdots}}}
\end{aligned}
$$

Lambert W function 恒等式

递推关系III：$-\frac{1}{e} < x < e, \quad |W_0(x)| < 1$

$$W_0(x) = \cfrac{x}{\exp \cfrac{x}{\exp \cfrac{x}{\cdots}}}$$

Lambert W function 应用

Applications range widely, from combinatorics to delay differential equations, to falling objects subject to linear drag, to the evaluation of the numerical constant in Wien's displacement law, to quantum statistics, to constructing the *tortoise* coordinate for Schwarzschild black holes, to GUP correction of thermodynamics of black hole, etc.

主要依据

$$Y = Xe^X \quad \Leftrightarrow \quad X = W(Y)$$

Lambert W function 应用

1. 求解超越方程

$$p^x = ax + b, \qquad p > 0, \quad a \neq 0$$

解答：做代换 $-t = x + \frac{b}{a}$，于是原方程化简为 $tp^t = R \equiv -\frac{1}{a}p^{-b/a}$，进一步写为 $te^{t\ln p} = R$，然后两边同乘以因子 $\ln p$，从而得到 $t\ln p = W(R\ln p)$，于是原方程的解写为：

$$x = -\frac{b}{a} - \frac{W\left(-\frac{\ln p}{a}p^{-b/a}\right)}{\ln p} \qquad \blacksquare$$

Lambert W function 应用

2. Wien's displacement law

$$\lambda_{\max} T = b$$

where b is Wien's displacement constant, $\lambda_{\max}$ is wavelength and T is absolute temperature of black-body radiation.

Lambert W function 应用

This law was proposed by Wien in 1893 from general thermodynamic arguments. The spectral distribution of black-body radiation is described by $\rho(\lambda, T)$. Let $\rho(\lambda, T)d\lambda$ is the power emitted in a wavelength interval $d\lambda$ per unit area from a black body at absolute temperature T. The wavelength $\lambda_{\max}$ at which ρ is a maximum obeys Wien's displacement law. Once Planck's spectral distribution law is known, Wien's law can be deduced and the value of b determined.

Lambert W function 应用

The Planck spectral distribution law is

$$\rho(\lambda, T) = \frac{8\pi hc/\lambda^5}{e^{hc/(\lambda kT)} - 1}$$

The value of λ for which this function is a maximum can be obtained by solving

$$\frac{\partial \rho(\lambda, T)}{\partial \lambda} = 0$$

Lambert W function 应用

This leads to the equation

$$-5e^{\frac{hc}{\lambda kT}} + 5 + \frac{hc}{\lambda kT} \cdot e^{\frac{hc}{\lambda kT}} = 0$$

One sets $x = hc/(\lambda kT)$, we can obtain the transcendental equation

$$(x-5)e^{x} = -5$$

Lambert W function 应用

This transcendental equation have solutions

$$x = \begin{cases} 0, & \text{trivial} \\ 5 + W(-5e^{-5}), & \text{nontrivial} \end{cases}$$

Therefore Wien's law is obtained with a new expression for Wien's displacement constant:

$$b = \frac{hc/k}{5 + W(-5e^{-5})} = 2.893 \times 10^{-3}\text{mK} \qquad \blacksquare$$

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练习

- 求解方程：(其中a, b, z均为正实数)
  i). $3^x = 4x$,　ii). $x^x = z$,　iii) $x \log_b x = a$.

- 完成下列积分：

$$\int_0^\infty \frac{W(x)}{x\sqrt{x}}dx, \quad \int_0^\infty W\left(\frac{1}{x^2}\right)dx$$